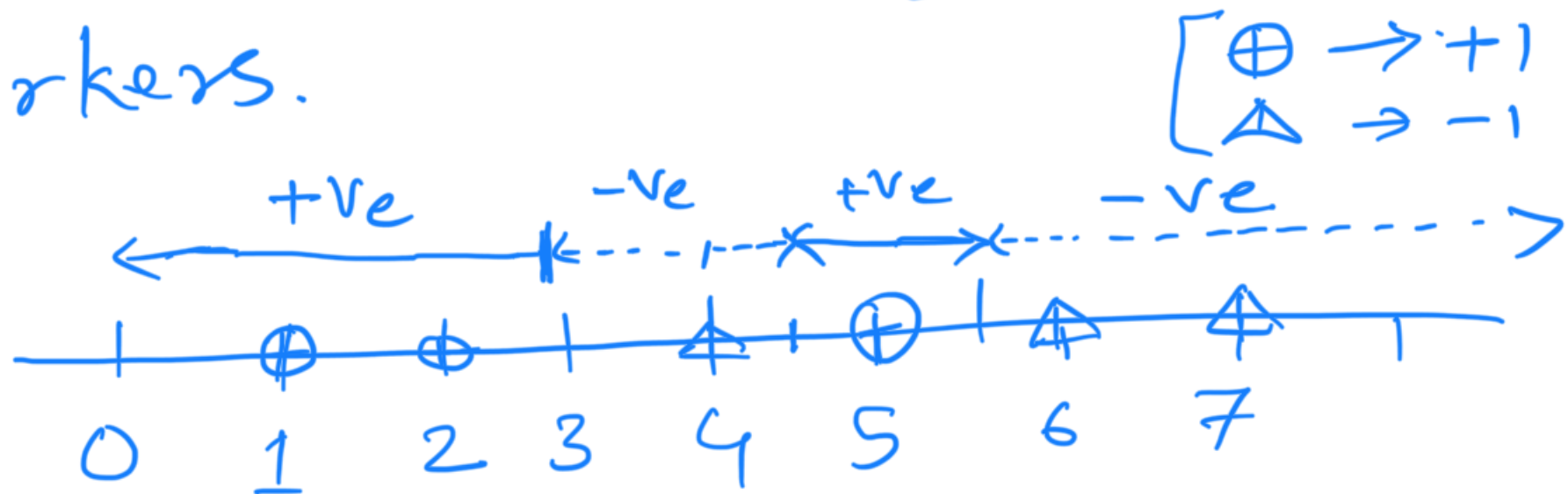


Question - Given One-dim. training points and their labels: -

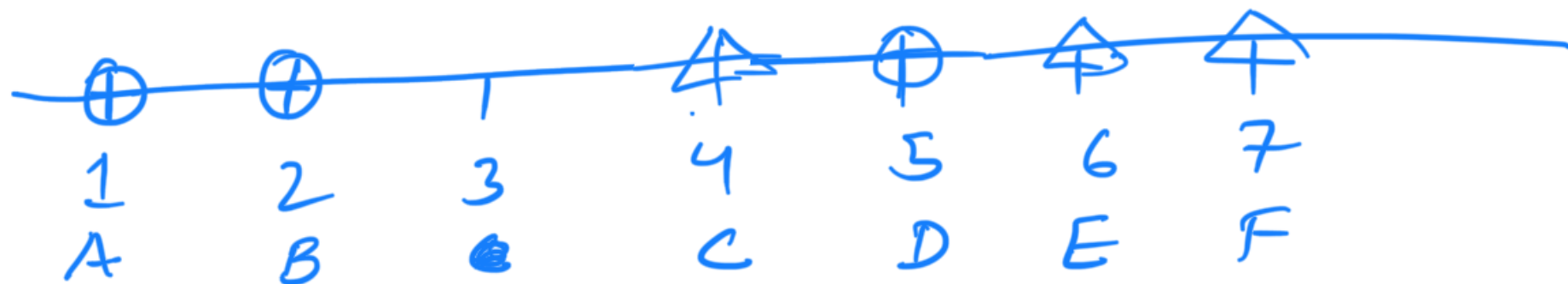
$(1, +1)$, $(2, +1)$, $(4, -1)$, $(5, +1)$, $(6, -1)$
 $(7, -1)$

(a) Show the points using appropriate markers.



(b) Show the regions where test samples will be classified as +ve. if we use 1-NN.

(c) Compute training accuracy in percentage if $k=3$ is used in k -NN.



~~Nearest~~ 3-Nearest points

Assign label
Actual label

A → A, B, C
+ + -

✓ +1 +1

B → B, A, C
+ + -

✓ +1 +1

✓ C → C, D, B (Rule: Tie Break) X +1 -1
- + +

D → D, C, E

X -1 +1

$$E \rightarrow \begin{matrix} + & - & - \\ \underline{E} & , D, \underline{F} \\ - & + & - \end{matrix}$$

$$\checkmark - 1 - 1$$

$$F \rightarrow \begin{matrix} & & \\ \underline{F}, \underline{E}, \underline{D} \\ - & - & + \end{matrix}$$

$$\checkmark - 1 - 1$$

$$\begin{aligned} \text{Training Error} &= \frac{2}{6} \times 100 \\ &= \frac{1}{3} \times 100 \\ &= 33.3\% \end{aligned}$$

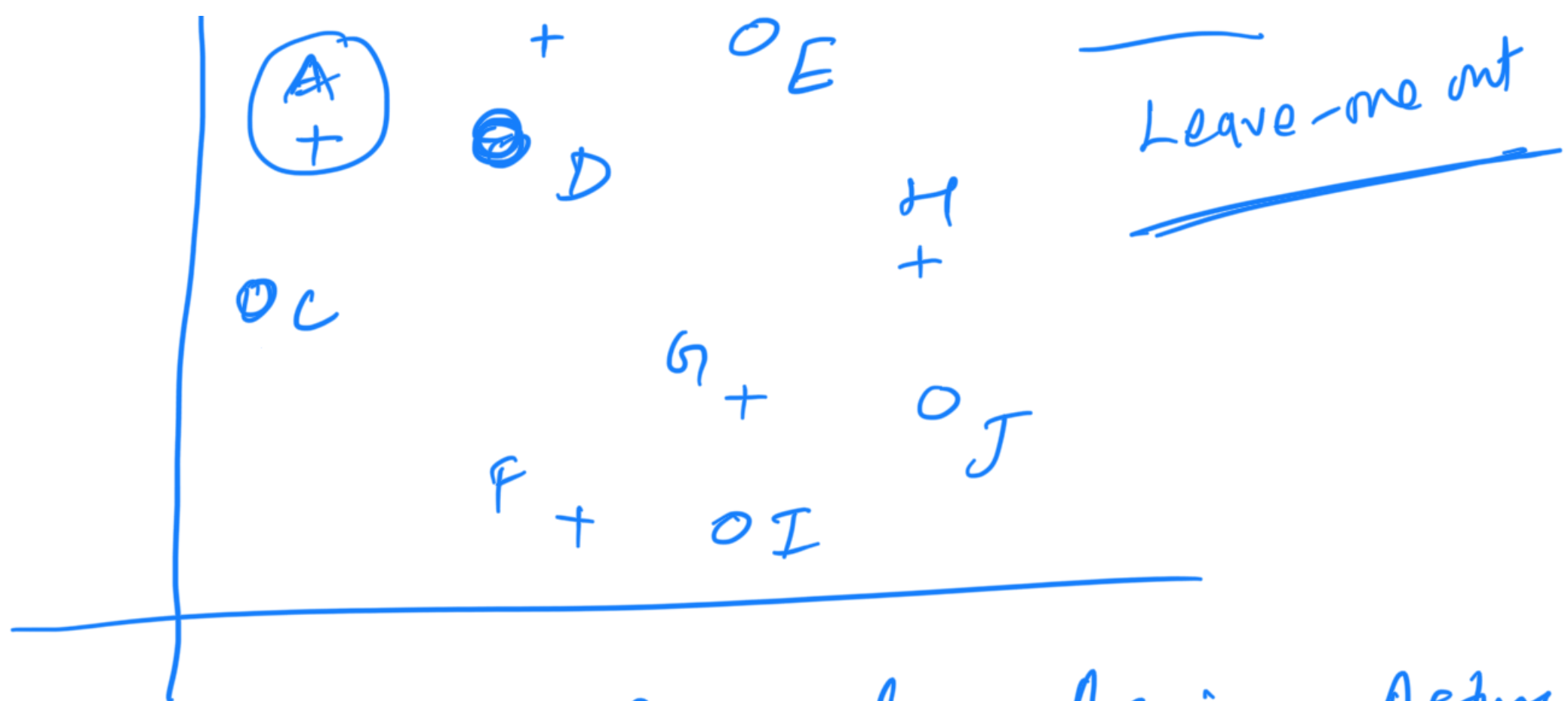
$$\begin{aligned} \text{Training Accuracy} &= \frac{4}{6} \times 100 \\ &= \frac{2}{3} \times 100 \\ &= \underline{66.7\%} \end{aligned}$$

Q2

1

B

1-NN



	Nearest Sample	Assign label	Actual label
A	C	0	+
F	I	0	+
J	G	+	0

1-NN always gives 100%
Training Accuracy.